

2.4 Nitrogen & Sulfur

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Medium

Time allowed: 70

Score: /51

Percentage: /100

Question 1a

Nitrogen and oxygen will react together under conditions of high temperature to give oxides of nitrogen.

i)

Draw the dot-and-cross diagram of a molecule of nitrogen.

[2]

ii)

Suggest and explain why high temperatures are required for oxides of nitrogen to form.

[3]

[5 marks]

Question 1b

i)

Write an equation for a possible reaction between nitrogen and oxygen.

[1]

ii)

The bond enthalpy of nitrogen is $+1000 \text{ kJ mol}^{-1}$.

Suggest the bond enthalpy of oxygen.

[1]

Possible bond enthalpy of oxygen kJ mol^{-1}

[2 marks]

Question 1c

i)

State **two** situations, **one** natural and **one** as a result of human activities, in which nitrogen and oxygen react together.

[2]

ii)

State **two** environmental effects of the presence of nitrogen oxides in the atmosphere.

[2]

[4 marks]**Question 1d** N_2O_5 is a rare and unstable oxide of nitrogen.State the systematic name of N_2O_5 .**[1 mark]**

Question 2a

Nitrogen oxides in the atmosphere are homogeneous catalysts in the formation of acid rain.

What is meant by the following terms?

catalyst

homogeneous

[3 marks]

Question 2b

State a major source of nitrogen oxides in the atmosphere, explaining how they are formed.

[2 marks]

Question 2c

Use equations to describe the chemical role played by nitrogen oxides in the formation of acid rain.

[3 marks]

Question 2d

Use the following axes in Fig. 1.1 to draw a fully labelled reaction pathway diagram showing the effect of a catalyst on an exothermic reaction.

Label the ΔH and E_a values.

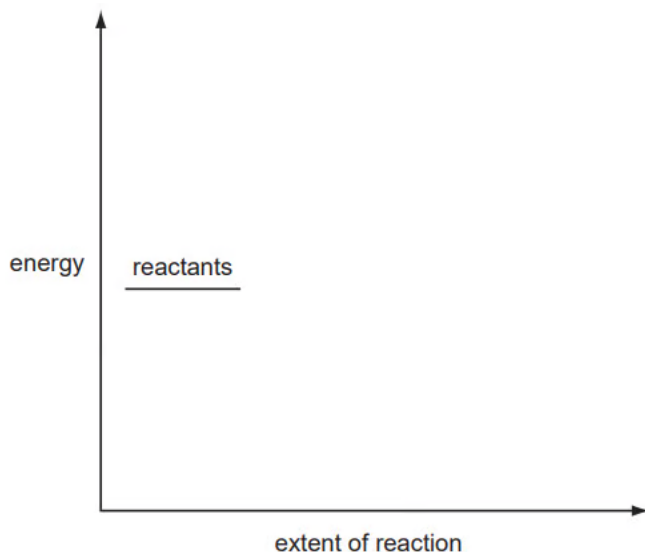


Fig. 1.1

[3 marks]

Question 3a

Ammonia, NH_3 , is a colourless, pungent-smelling gas that has been known to man from the beginning of recorded time. Now ammonia is synthesised from its elements in the Haber process.

i)

Write an equation for this process.

[1]

ii)

State the **three** usual operating conditions of the Haber process.

[1]

[2 marks]

Question 3b

Ammonia can act as a Brønsted–Lowry base.

Explain why.

[2 marks]

Question 3c

1.50 dm³ of ammonia gas were dissolved in water to form 300 cm³ of aqueous alkali at room temperature and pressure.

i)

Calculate how many moles of NH₃(g) were dissolved.

Show your working.

[1]

ii)

Write the equation for the neutralisation of aqueous ammonia by dilute sulfuric acid.

[1]

iii)

Calculate the volume of 0.80 mol dm⁻³ sulfuric acid that is required to neutralise the 300 cm³ of aqueous ammonia.

Show your working.

[2]

[4 marks]

Question 3d

i)

In the boxes below, draw diagrams to show the shapes of an ammonia molecule and an ammonium ion.
Clearly show the bond angles on your diagrams.

[4]

Ammonia	Ammonium ion

ii)

State the name of the shape of an ammonia molecule and an ammonium ion.

Ammonia molecule

Ammonium ion

[2]

[6 marks]

Question 3e

Ammonia gas can also be prepared by heating a mixture of ammonium chloride and calcium hydroxide.

Give the equation for this reaction.

[2 marks]

Question 4a

The high temperatures reached in an internal combustion engine cause nitrogen and oxygen from the air to react. Nitrogen monoxide is formed.

i)

Give the equation for this reaction.

[1]

ii)

Show, using oxidation numbers, that nitrogen has been oxidised.

[1]

iii)

Nitrogen monoxide will react further with oxygen to form nitrogen dioxide.

Give the equation for this oxidation reaction.

[1]

[3 marks]

Question 4b

Nitrogen oxides play a role in the formation of photochemical smog. Explain how.

[3 marks]**Question 4c**

The brown haze associated with photochemical smog is typically worse in cities in the afternoons of hot sunny days.

Suggest why.

[3 marks]**Question 4d**

Catalytic converters are fitted to exhaust systems to reduce the pollutants emitted by car engines. Nitrogen oxide, NO, and carbon monoxide, CO are removed in a catalytic converter, forming nitrogen and carbon dioxide, which are less harmful to human health.

i)

Give the balanced symbol equation for the removal of NO and CO.

[1]

ii)

State, using oxidation numbers, what is acting as a reducing agent in this reaction.

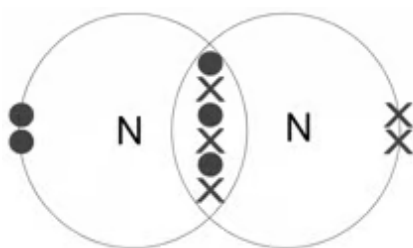
[2]**[3 marks]**

Model Answers: Medium

1a

a)

i) The dot-and-cross diagram of a molecule of nitrogen:



- 3 shared pairs of electrons between two nitrogen atoms; [1 mark]
- 2 non-bonded electrons on each nitrogen atom; [1 mark]

ii) High temperatures are required for oxides of nitrogen to form because:

- Nitrogen is very unreactive; [1 mark]
- (Because) it requires a lot of energy to break the (triple) bond; [1 mark]
- It is a non-polar molecule so is not attracted to other molecules; [1 mark]

[Total: 5 marks]

- Each nitrogen atom has 5 electrons, so in order for each atom to gain a full outer shell, it needs to share 3 electrons
 - A triple bond is formed
- After you have drawn the dot-and-cross diagram, ensure you check that each nitrogen atom has 8 electrons
- High temperatures are needed for nitrogen and oxygen to react, due to the lack of reactivity of nitrogen, so part (ii) is essentially asking you to state and explain the reactivity of nitrogen

1b

b)

i) A possible equation for a reaction between nitrogen and oxygen is:



OR



ii) A possible value for the bond enthalpy of oxygen is:

- A value below 1000 and above 0 kJ mol^{-1} ; [1 mark]

[Total: 2 marks]

- You can give the equation for the formation of either nitrogen monoxide, NO, or nitrogen dioxide, NO₂
- The bond enthalpy of oxygen will be less than the bond enthalpy of nitrogen because the nitrogen-nitrogen triple bond will require more energy to break than the oxygen-oxygen double bond
- **Remember:** The value must be positive as bond enthalpies are always endothermic

1c

c)

i) **Two** situations in which nitrogen and oxygen react together are:

- Natural: Lightning; [1 mark]
- As a result of human activity: In an engine / combustion of fuels; [1 mark]

ii) **Two** environmental effects of the presence of nitrogen oxides in the atmosphere are:

- (NO_x produces) acid rain; [1 mark]
- (NO_x) forms (photochemical) smog; [1 mark]

[Total: 4 marks]

- You could give a specific example of the combustion of fuels
- This can occur in a car engine or in an aeroplane engine

1d

d) The systematic name of N_2O_5 is:

- Nitrogen(V) oxide; [1 mark]

[Total: 1 mark]

- There are many oxides of nitrogen that exist, some are less stable than others
- The main two oxides of nitrogen that you will come across are nitrogen monoxide, NO , and nitrogen dioxide, NO_2
- To give the systematic name, you need to find the oxidation number of nitrogen
 - There are 5 O atoms, each with an oxidation number of -2, so -10 due to oxygen
 - Overall, the oxidation number of the compound is 0
 - The oxidation number of the 2 N atoms combined must be +10, so each N must be +5
- The Roman numeral after nitrogen shows the oxidation number of nitrogen in this compound

2a

a) The meanings of the terms **catalyst** and **homogenous** are:

Catalyst: Any **two** from:

- Speeds up / increases (NOT alters or changes) the rate of a reaction; [1 mark]
- Lowers energy barrier / E_a
OR
Offers a lower energy pathway / mechanism; [1 mark]
- Is not used up or remains unchanged
OR
Does not alter its mass / concentration
OR
Does not appear in stoichiometric equation
OR
Is regenerated; [1 mark]

Homogenous:

- (The catalyst and reactants are) in the same phase / state; [1 mark]

[Total: 3 marks]

- **Careful:** You would not be awarded a mark if you said a catalyst alters or changes the rate of reaction as it is not specific enough, you must imply that the rate increases
- These are definitions that you need to invest time in learning as they are easy marks to pick up

2b

b) A major source of nitrogen oxides in the atmosphere, explaining how they are formed is:

- Car exhausts / engines / aeroplanes / lightning / burning fuels / power stations; [1 mark]
- (They are formed as) nitrogen / N_2 reacts with oxygen / O_2 ; [1 mark]

[Total: 2 marks]

- This is a common question so ensure that you know how nitrogen oxides are formed both naturally and through human activities

2c

c) The equations to describe the chemical role played by nitrogen oxides in the formation of acid rain are:

Any **three** from:

- $NO_2 + SO_2 \rightarrow NO + SO_3$; [1 mark]
- $NO + \frac{1}{2} O_2 \rightarrow NO_2$; [1 mark]
- $SO_3 + H_2O \rightarrow H_2SO_4$; [1 mark]
- $4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

OR

**[Total: 3 marks]**

- Nitrogen oxides can form acid rain directly:
 - $4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

- Alternatively, nitrogen oxides can act as a catalyst in the formation of acid rain
- When SO_2 , another atmospheric pollutant, is oxidised, it forms SO_3 which reacts with rainwater to form dilute sulfuric acid:
 - $\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$
- NO_2 catalyses the oxidation of SO_2 to SO_3 :
 - $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$
- The formed NO gets oxidised to regenerate NO_2 :
 - $\text{NO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$

2d

d) The fully labelled reaction pathway diagram showing the effect of a catalyst on an exothermic reaction is:



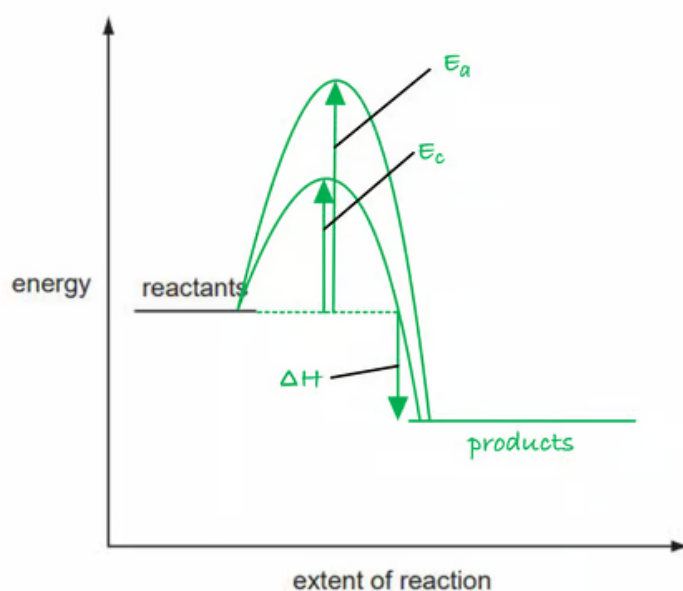
- $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$; [1 mark]
 - $4\text{NO}_2 + 2\text{H}_2\text{O} + \text{O}_2 \rightarrow 4\text{HNO}_3$
- OR
- $3\text{NO}_2 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3 + \text{NO}$; [1 mark]

[Total: 3 marks]

- Nitrogen oxides can form acid rain directly:
 - $4\text{NO}_2 + 2\text{H}_2\text{O} + \text{O}_2 \rightarrow 4\text{HNO}_3$
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- NO_2 catalyses the oxidation of SO_2 to SO_3 :
 - $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$
- The formed NO gets oxidised to regenerate NO_2 :
 - $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$

2d

d) The fully labelled reaction pathway diagram showing the effect of a catalyst on an exothermic reaction is:



- ΔH shown as negative; [1 mark]
- The catalysed and uncatalysed E_a correctly labelled; [1 mark]

- $E_a > E_c$; [1 mark]

[Total: 3 marks]

- If you drew this as an endothermic reaction, i.e. the products higher than the reactants, you could have still scored the final two marks
- Make sure that your arrows are pointing in the correct direction and that you do not use double-headed arrows
- **Careful:** You need to make sure that you draw both the catalysed and uncatalysed reaction pathway diagram in order to show the effect of a catalyst

3a

a)

i) An equation for this process is:

- $N_2 + 3H_2 \rightleftharpoons 2NH_3$; [1 mark]

ii) The **three** usual operating conditions of the Haber process are:

- Pressure: 100 atm or over
AND
Temperature: 400 - 500°C
AND
Catalyst: iron; [1 mark]

[Total: 2 marks]

- You should know the conditions and equation for the Haber process
- An explanation of the reasons why these conditions are used is a common question that should be answered in terms of the compromises between equilibrium yield, rate of reaction and cost

3b

b) Ammonia can act as a Brønsted–Lowry base because:

- It is able to accept a proton / H^+ ion; [1 mark]
- To form the ammonium ion / NH_4^+ ; [1 mark]

[Total: 2 marks]

- **Remember:** Acids are proton donors, bases are proton acceptors
- When ammonia accepts a proton, the ammonium ion is formed:
 - $\text{NH}_3(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{NH}_4^+(\text{aq})$

3c

c)

i) The number of moles of $\text{NH}_3(\text{g})$ that were dissolved is:

- Moles $\text{NH}_3 = 1.5 \div 24 = 0.0625$ (mol); [1 mark]

ii) The equation for the neutralisation of aqueous ammonia by dilute sulfuric acid is:

- $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$; [1 mark]

iii) The volume of 0.80 mol dm^{-3} sulfuric acid that is required to neutralise the 300 cm^3 of aqueous ammonia is:

- Moles of H_2SO_4 required = 0.03125 (mol); [1 mark]
- Vol of 0.80 mol dm^{-3} H_2SO_4 required = $(0.03125 \times 1000) / 0.80 = 39 / 39.1$ (cm^3); [1 mark]

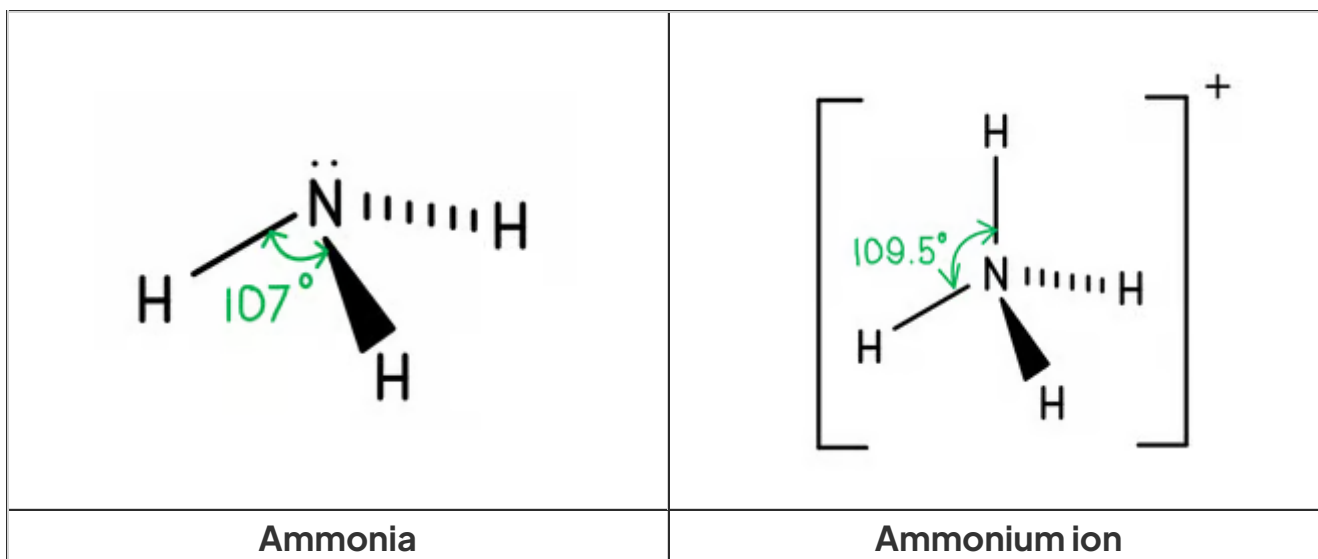
[Total: 4 marks]

- You need to use the molar gas volume, which is 24.0 dm^3 at RTP, to find the number of moles of ammonia
- You are given this value in an exam
- **Remember:** The neutralisation of ammonia and acid produces ammonium salt but NO water
- Use the balanced symbol equation to work out the number of moles of sulfuric acid
- The volume is calculated by moles $\div$ conc but this will need to be multiplied by 1000 to give the value in cm^3

3d

d)

i) Diagrams to show the shapes of an ammonia molecule and an ammonium ion. and the bond angles:



- Correct diagram of ammonia molecule; [1 mark]
- Correct bond angle of 107° between N-H bonds; [1 mark]
- Correct diagram of ammonium ion; [1 mark]
- Correct bond angle of 109.5° between N-H bonds; [1 mark]

ii) The name of the shapes of an ammonia molecule and an ammonium ion are:

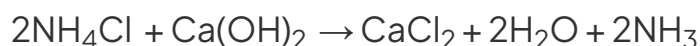
- Ammonia molecule: Pyramidal; [1 mark]
- Ammonium ion: Tetrahedral; [1 mark]

[Total: 6 marks]

- Ammonia has a lone pair of electrons which pushes the bonded pairs of electrons together, with a bond angle of 107° and a pyramidal shape
- When ammonium ions are formed, the lone pair of electrons on the ammonia molecule is used to form a dative covalent bond
 - The 4 bonded pairs of electrons repel each other equally giving the ammonium ion a tetrahedral shape with a bond angle of 109.5°

3e

e) The equation for the reaction of ammonium chloride and calcium hydroxide is:



- Correct reactants and products; [1 mark]

- Correct balancing; [1 mark]

[Total: 2 marks]

- Ammonia gas can be prepared from an ammonium salt and a base in an acid-base reaction:
- The salt and base are mixed and then heated
- NH_4^+ acts as an acid (proton donor) and OH^- acts as a base (proton acceptor)

4a

a)

i) The equation for this reaction is:

- $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$; [1 mark]

ii) To show that nitrogen has been oxidised:

- The oxidation number of nitrogen has increased from 0 (in N_2) to +2 (in NO) (and has

- You should know the conditions and equation for the Haber process
- An explanation of the reasons why these conditions are used is a common question that should be answered in terms of the compromises between equilibrium yield, rate of reaction and cost

3b

b) Ammonia can act as a Brønsted–Lowry base because:

- It is able to accept a proton / H^+ ion; [1 mark]
- To form the ammonium ion / NH_4^+ ; [1 mark]

[Total: 2 marks]

- **Remember:** Acids are proton donors, bases are proton acceptors
- When ammonia accepts a proton, the ammonium ion is formed:
 - $\text{NH}_3(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{NH}_4^+(\text{aq})$

3c

c)

i) The number of moles of $\text{NH}_3(\text{g})$ that were dissolved is:

- Moles $\text{NH}_3 = 1.5 \div 24 = 0.0625$ (mol); [1 mark]

ii) The equation for the neutralisation of aqueous ammonia by dilute sulfuric acid is:

- $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$; [1 mark]

iii) The volume of 0.80 mol dm^{-3} sulfuric acid that is required to neutralise the 300 cm^3 of aqueous ammonia is:

- Moles of H_2SO_4 required = 0.03125 (mol); [1 mark]
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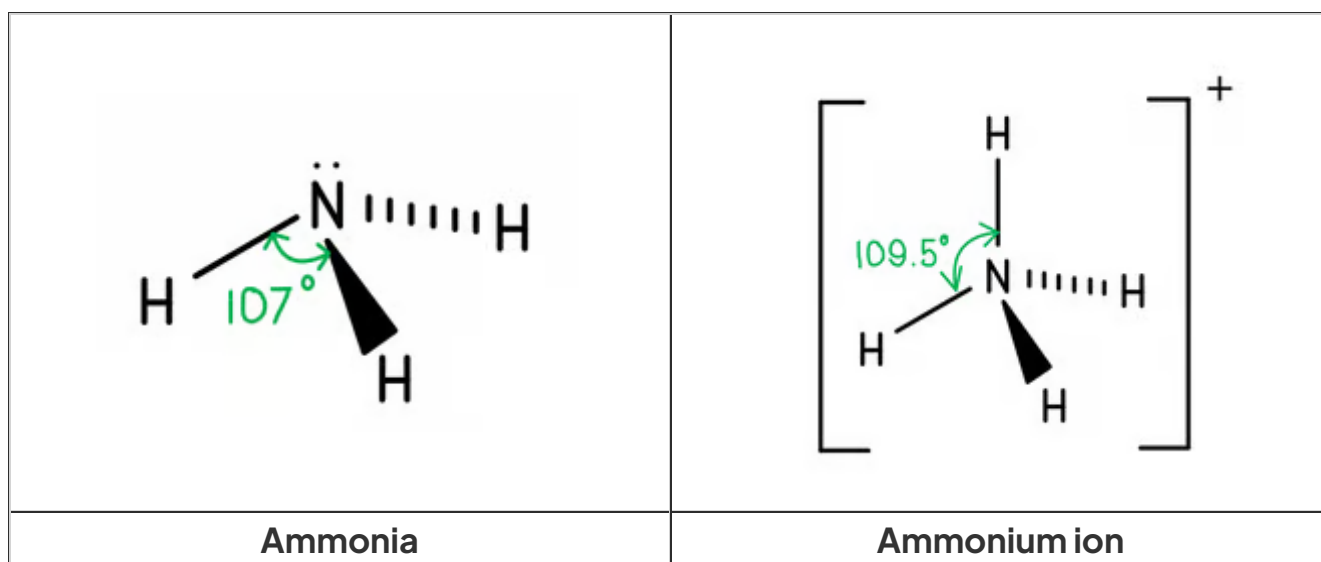
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- You are given this value in an exam
- **Remember:** The neutralisation of ammonia and acid produces ammonium salt but NO water
- Use the balanced symbol equation to work out the number of moles of sulfuric acid
- The volume is calculated by $\text{moles} \div \text{conc}$ but this will need to be multiplied by 1000 to give the value in cm^3

3d

d)

i) Diagrams to show the shapes of an ammonia molecule and an ammonium ion. and the bond angles:



- Correct diagram of ammonia molecule; [1 mark]
- Correct bond angle of 107° between N-H bonds; [1 mark]
- Correct diagram of ammonium ion; [1 mark]
- Correct bond angle of 109.5° between N-H bonds; [1 mark]

ii) The name of the shapes of an ammonia molecule and an ammonium ion are:

- Ammonia molecule: Pyramidal; [1 mark]
- Ammonium ion: Tetrahedral; [1 mark]

[Total: 6 marks]

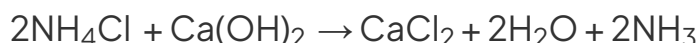
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together, with a bond angle of 107° and a pyramidal shape

- When ammonium ions are formed, the lone pair of electrons on the ammonia molecule is used to form a dative covalent bond
 - The 4 bonded pairs of electrons repel each other equally giving the ammonium ion a tetrahedral shape with a bond angle of 109.5°

3e

e) The equation for the reaction of ammonium chloride and calcium hydroxide is:



- Correct reactants and products; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- Ammonia gas can be prepared from an ammonium salt and a base in an acid-base reaction:
- The salt and base are mixed and then heated
- NH_4^+ acts as an acid (proton donor) and OH^- acts as a base (proton acceptor)

4a

a)

i) The equation for this reaction is:

- $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$; [1 mark]

ii) To show that nitrogen has been oxidised:

- The oxidation number of nitrogen has increased from 0 (in N_2) to +2 (in NO) (and has therefore been oxidised); [1 mark]

iii) The equation for the further oxidation of nitrogen monoxide is:

- $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$; [1 mark]

[Total: 3 marks]

- You must know the equations for the formation of NO and NO_2

- When explaining whether a species has been oxidised or reduced using oxidation numbers, always state what the oxidation number is of the element before and after the reaction
- **Remember:** An increase in oxidation number shows oxidation, and a decrease shows reduction

4b

b) Nitrogen oxides play a role in the formation of photochemical smog by:

- Nitrogen oxides react with unburnt hydrocarbons / VOCs; [1 mark]
- In the presence of light / sunlight; [1 mark]
- Forming PAN / peroxyacetyl nitrate / $\text{CH}_3\text{CO}_3\text{NO}_2$ (found in smog); [1 mark]

[Total: 3 marks]

- You do not need to be able to give the equations for this reaction
- PAN is just one of the pollutants found in photochemical smog, you do not need to know any of the other substances
- The reaction is a photochemical reaction as light is needed for it to occur

4c

c) The brown haze associated with photochemical smog is typically worse in cities in the afternoons of hot sunny days because:

- There are more cars, therefore more emissions in cities; [1 mark]
- The volume of PAN / VOCs increases during the day; [1 mark]
- There is lots of (sun)light (energy) present; [1 mark]

[Total: 3 marks]

- The 'suggest' command word means that you may not have studied this but you should be able to apply your knowledge to answer it
- There are three marks available, so you need to explain why smog is worse in
 1. cities
 2. afternoons
 3. on hot sunny days

4d

d)

i) The balanced symbol equation for the removal of NO and CO is:



ii) The species acting as the reducing agent in this reaction is:

- Carbon monoxide / CO; [1 mark]
- As it reduces N from +2 (in NO) to 0 (in N₂)

OR

Its oxidation number increases from +2 (in CO) to +4 (in CO₂); [1 mark]

[Total: 3 marks]

- A reducing agent is one which reduces another species, so is itself oxidised
- You can show which is the reducing agent by either showing, using oxidation numbers, which species is oxidised or that N is reduced